

### Module A (Lec 8): High-Dimensional Newton

**Newton step (linear solver).** Let  $g_k = \nabla f(x^{(k)})$  and  $F_k = \nabla^2 f(x^{(k)})$ . Compute  $d_k$  from

$$F_k d_k = -g_k, \quad x^{(k+1)} = x^{(k)} + d_k.$$

- **Never form  $F_k^{-1}$  explicitly.** Use Cholesky (if  $F_k \succ 0$ ) or iterative solvers (e.g. Conjugate Gradient).
- **Theorem 9.1 (Local Second Order Convergence)** Suppose that  $f \in C^3$  and  $x^* \in \mathbb{R}^n$  is a point such that  $\nabla f(x^*) = 0$  and  $F(x^*)$  is invertible. Then, for all  $x^{(0)}$  sufficiently close to  $x^*$ , Newton's method is well-defined for all  $k$  and converges to  $x^*$  with an order of convergence at least 2.  $\square$

#### Second-Order Optimality Conditions

- **SONC (Second-Order Necessary Condition):** If  $x^*$  is a local minimizer, then  $\nabla f(x^*) = 0$  AND  $\nabla^2 f(x^*) \succeq 0$  (PSD).
- **SOSC (Second-Order Sufficient Condition):** If  $\nabla f(x^*) = 0$  AND  $\nabla^2 f(x^*) \succ 0$  (PD), then  $x^*$  is a strict local minimizer.

### Module A (lec 9): Levenberg–Marquardt (L–M) Modification

**Theorem 9.2 (Newton Descent Direction)** Let  $\{x^{(k)}\}$  be the sequence generated by Newton's method for minimizing a given objective function  $f(x)$ . If the Hessian  $F(x^{(k)}) \succ 0$  and  $g^{(k)} = \nabla f(x^{(k)}) \neq 0$ , then the search direction

$$d^{(k)} = -F(x^{(k)})^{-1} g^{(k)} = x^{(k+1)} - x^{(k)}$$

from  $x^{(k)}$  to  $x^{(k+1)}$  is a descent direction for  $f$  in the sense that there exists an  $\bar{\alpha} > 0$  such that for all  $\alpha \in (0, \bar{\alpha})$ ,

$$f(x^{(k)} + \alpha d^{(k)}) < f(x^{(k)}).$$

$\square$

**Idea:** enforce a descent direction even when  $F_k \not\succ 0$ .

$$x^{(k+1)} = x^{(k)} - (F_k + \mu_k I)^{-1} g_k, \quad \mu_k \geq 0.$$

- **Eigenvalue shift:** eigenvalues of  $F_k + \mu_k I$  are  $\lambda_i(F_k) + \mu_k$ . Large enough  $\mu_k$  makes it SPD.
- **Interpolating Behavior:**  $\mu_k \rightarrow 0 \Rightarrow$  Newton;  $\mu_k \rightarrow \infty \Rightarrow$  small-step gradient.
- Often combine with line search:

$$x^{(k+1)} = x^{(k)} - \alpha_k (F_k + \mu_k I)^{-1} g_k.$$

### Practice: Exercise 9.8 (Modified Newton + exact line search on a quadratic)

Consider

$$x^{(k+1)} = x^{(k)} - \alpha_k F(x^{(k)})^{-1} g^{(k)}, \quad \alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} - \alpha F(x^{(k)})^{-1} g^{(k)}).$$

Apply it to

$$f(x) = \frac{1}{2} x^\top Q x - x^\top b, \quad Q = Q^\top \succ 0.$$

Does it reach  $x^*$  in one step from any  $x^{(0)}$ ?

## Module B (Ch. 11): Quasi-Newton Methods — Preview

### Form of the Family:

$$d_k = -H_k g_k, \quad x^{(k+1)} = x^{(k)} + \alpha_k d_k, \quad \alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} + \alpha d_k).$$

- $H_k$  approximates  $F_k^{-1}$  using **only** function/gradient info (no Hessian factorization each step).
- **Descent condition:**  $g_k^\top H_k g_k > 0$  (clean sufficient condition:  $H_k \succ 0$ ).

### Practice: Exercise 11.3 (Exact line search step size on a quadratic)

We run

$$d^{(k)} = -H_k g^{(k)}, \quad x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)}, \quad \alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} + \alpha d^{(k)}),$$

with

$$f(x) = \frac{1}{2} x^\top Q x - x^\top b + c, \quad Q = Q^\top \succ 0.$$

- (a) Find a closed form for  $\alpha_k$  in terms of  $Q$ ,  $g^{(k)}$ , and  $d^{(k)}$  (and hence  $H_k$ ).

- (b) Give a sufficient condition on  $H_k$  for  $\alpha_k$  to be positive (assume  $g^{(k)} \neq 0$ ).

## Module C: Armijo Line Search (Backtracking)

Pick a descent direction  $d_k$  with  $g_k^\top d_k < 0$  and choose  $\alpha_k$  so that

$$f(x^{(k)} + \alpha_k d_k) \leq f(x^{(k)}) + c \alpha_k g_k^\top d_k, \quad c \in (0, 1).$$

- Start  $\alpha = 1$ ; while Armijo fails, set  $\alpha \leftarrow \beta \alpha$  ( $\beta \in (0, 1)$ , e.g.  $\frac{1}{2}$ ).
- Terminates since  $f(x^{(k)} + \alpha d_k) = f(x^{(k)}) + \alpha g_k^\top d_k + O(\alpha^2)$  and  $g_k^\top d_k < 0$ .